Advanced high school · geometry

Rotations and complex numbers

Lesson and method

Before you begin

Algebraic form, modulus and coordinates.

Ideas and method

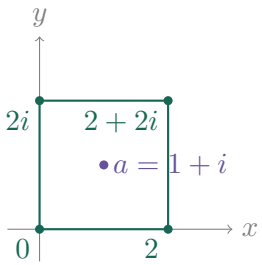
The complex number $z = x$+iy represents (x,y). Multiplying by i gives $iz = -y + ix$, which represents (−y,x): a counterclockwise quarter-turn about the origin.

Multiplication by −1 gives a half-turn, and by −i a clockwise quarter-turn. They preserve distances from the origin because $|i| = 1$.

For rotation about centre a, translate to the origin, rotate and translate back: $z' = a + i(z - a)$. This counterclockwise rotation preserves distances between all points.

Worked example

With $a = 1 + i$ and $z = 3 + 2i, z - a = 2 + i$ and $i(z - a) = -1 + 2i$. The image is $z' = 3i$. Relative vectors (2,1) and (−1,2) are perpendicular and both have norm $\sqrt{5}$.



Explore

In the complex-number lab, move z and compare z with iz. Check equal moduli and perpendicular vectors.